

Problem 1: Put 6 counters in piles of 4, 1 and 1. A move goes like this: take one counter from every pile, and put those counters together as one new pile. Make one move. In row 1 of the table, write the sizes of your piles, biggest first. Keep making moves and writing a row for each one, until a row is exactly the same as the row above it.

Move	Piles, biggest first
Start	4 1 1
1	
2	
3	
4	
5	
6	
7	
8	

Problem 2: Do the same from two more starts: all 6 counters in one pile, and six piles of 1. Use one table for each start.

Move	Piles, biggest first
Start	6
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	1 1 1 1 1 1
1	
2	
3	
4	
5	
6	
7	
8	

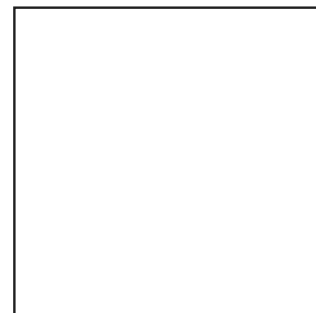
Problem 3: Do the same from two more starts: two piles of 3, and piles of 2, 2, 1 and 1.

Move	Piles, biggest first
Start	3 3
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	2 2 1 1
1	
2	
3	
4	
5	
6	
7	
8	

Problem 4: Fill in the table: for each of your five starts, how many moves it took until the piles stopped changing, and what the piles were then. Then build the end piles as towers standing side by side, biggest on the left, and draw the towers in the box.

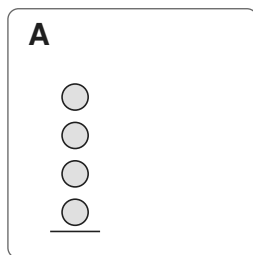
Start	Moves until it stopped	Piles at the end
4 1 1		
6		
1 1 1 1 1 1		
3 3		
2 2 1 1		



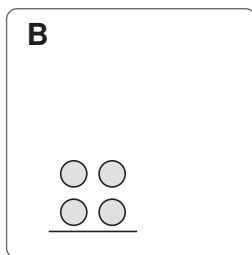
Problem 5: Take 4 counters in one pile. Make 6 moves and write a row for each move. Circle every row that is the same as row 1. Does this game ever stop changing?

Move	Piles, biggest first
Start	4
1	
2	
3	
4	
5	
6	

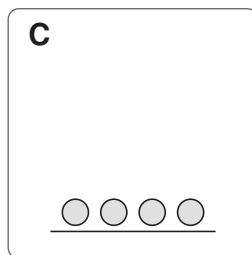
Problem 6: Here are all five ways to put 4 counters in piles, with letters. For each one, make the arrangement with your counters, make one move, and write the letter of the arrangement you get in the blank. Then answer: which letters does the game go round and round on? Which arrangement can you never get by a move from another one?



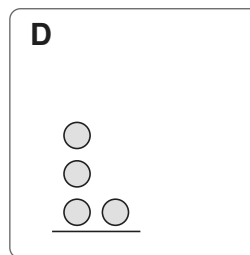
→ _____



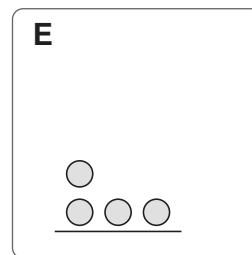
→ _____



→ _____



→ _____



→ _____

Goes round on: _____

Never reached by a move: _____

Problem 7: Take 5 counters in one pile and make 6 moves, writing each row. Then start with five piles of 1 and make 6 moves. In each table, circle the rows that repeat. Do both games go round on the same arrangements?

Move	Piles, biggest first
Start	5
1	
2	
3	
4	
5	
6	

Move	Piles, biggest first
Start	1 1 1 1 1
1	
2	
3	
4	
5	
6	

Same arrangements in both loops? _____

Problem 8: Take 7 counters in one pile. Make 8 moves and write each row. Then start from piles of 2, 2, 2 and 1 and make 7 moves. In each table, circle the rows that repeat. Do both games go round on the same arrangements? How many different arrangements are in the loop?

Move	Piles, biggest first
Start	7
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	2 2 2 1
1	
2	
3	
4	
5	
6	
7	

Same arrangements in both loops? _____

Arrangements in the loop: _____

Problem 9: For 3 counters, start with three piles of 1, make moves and write each row until a row is the same as the row above it. Then fill in the table for 3, 4, 5, 6 and 7 counters: does the game stop, or go round and round? If it goes round, how many different arrangements are in the loop?

Move	Piles, biggest first
Start	1 1 1
1	
2	
3	

Counters	Stops, or goes round?	Arrangements in the loop
3		
4		
5		
6		
7		

Problem 10: Look at your tables in Problems 4 and 9: the game with 3 counters stopped at 2 1, and the game with 6 counters stopped at 3 2 1. Build the stairs with towers of 3, 2 and 1. Make it one step taller by adding a tower of 4 beside the tower of 3, and count the counters. Write the number in the first blank. Write your prediction in the second blank: will the game with that many counters stop, or go round? Then test it: put that many counters in one pile, write the start in the table, and write each row until a row is the same as the row above it.

Counters in the next stairs: _____ Prediction: _____

Move	Piles, biggest first
Start	
1	
2	
3	
4	
5	
6	
7	
8	

Moves until it stopped: _____

Problem 11: Run 10 counters from two more starts: five piles of 2, and two piles of 5. Write each row until a row is the same as the row above it. Write how many moves each start takes.

Move	Piles, biggest first
Start	2 2 2 2 2
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	5 5
1	
2	
3	
4	
5	
6	
7	
8	

Moves from 2 2 2 2 2: _____

Moves from 5 5: _____

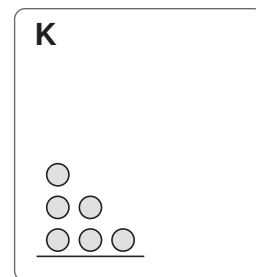
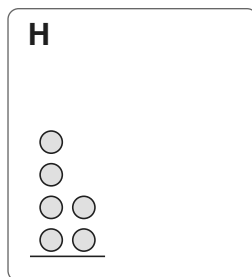
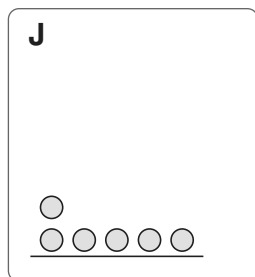
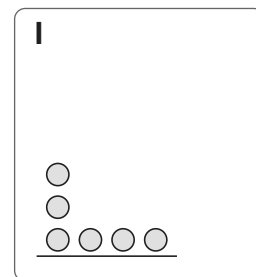
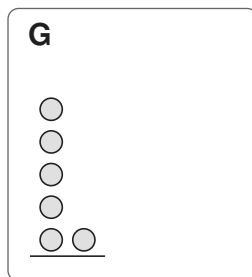
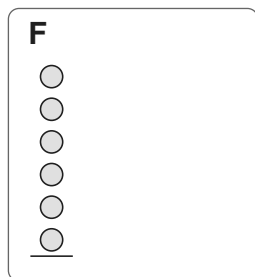
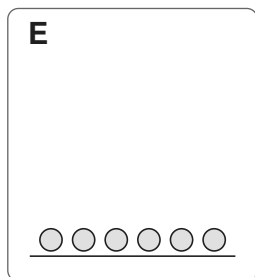
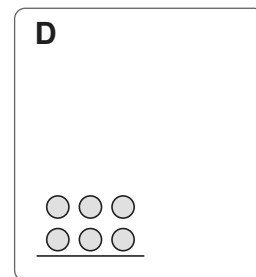
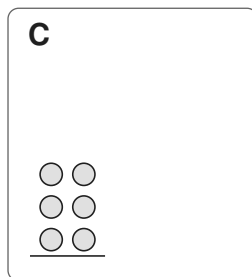
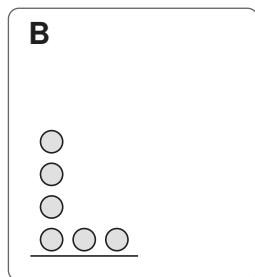
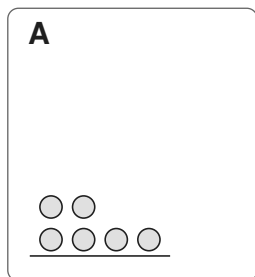
Problem 12: Now 8 counters. Put all 8 in one pile and make 8 moves, writing each row. Then put 8 counters in four piles of 2 and make 5 moves. In each table, circle the rows that repeat. Do the two games go round on the same arrangements?

Move	Piles, biggest first
Start	8
1	
2	
3	
4	
5	
6	
7	
8	

Move	Piles, biggest first
Start	2 2 2 2
1	
2	
3	
4	
5	

Same arrangements in both loops? _____

Problem 13: Here are all 11 ways to put 6 counters in piles, drawn as towers, with letters A to K. For each picture: make it with your counters, make one move, find the picture of the result, and draw an arrow from the first picture to the second. If the result is the same picture you started from, draw a small arrow that loops back to that picture. Then answer the three questions under the map.



Which pictures have no arrow pointing at them? _____

Which picture do all the arrows lead to in the end? _____

Starting at A, how many arrows do you follow to get there? _____

Problem 14: Here are all 22 ways to put 8 counters in piles. Share them out at your table. For each one, make moves and write the rows on a blank sheet until a row is one of the six loop arrangements from Problem 12. Write 4 next to the start if that row was 4 3 1, 3 3 2, 3 2 2 1 or 4 2 1 1, and 2 if it was 4 2 2 or 3 3 1 1. If after 6 moves you have reached neither, show your run to the adult. How many of the 22 go round on two?

8	_____	3 3 2	_____
7 1	_____	3 3 1 1	_____
6 2	_____	3 2 2 1	_____
6 1 1	_____	3 2 1 1 1	_____
5 3	_____	3 1 1 1 1 1	_____
5 2 1	_____	2 2 2 2	_____
5 1 1 1	_____	2 2 2 1 1	_____
4 4	_____	2 2 1 1 1 1	_____
4 3 1	_____	2 1 1 1 1 1 1	_____
4 2 2	_____	1 1 1 1 1 1 1 1	_____
4 2 1 1	_____		
4 1 1 1 1	_____		

Number that go round on two arrangements: _____

Problem 15: Build the stairs with 5 steps. Count the counters and write the number in the blank. Then put that many counters in five piles of 3. Make moves and write each row until a row is the same as the row above it. How many moves did it take?

Counters in the stairs with 5 steps: _____

Move	Piles, biggest first
Start	3 3 3 3 3
1	
2	
3	
4	
5	
6	

Moves until it stopped: _____